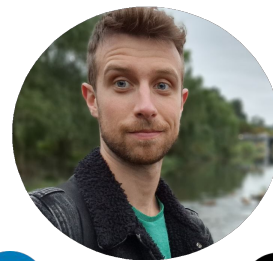


Edoardo Mosca

📍 Munich, Germany | ✉ edoardo.mosca@tum.de | 🌐 [EdoardoMosca.github.io](https://www.EdoardoMosca.github.io)

XAI & NLP Researcher, author of 10+ scientific publications, led 20+ ML projects, and awarded by the Germany Ministry for research. Strong background in Mathematics + PhD in Computer Science with experience in the industry, academia, and as a startup co-founder.



EXPERIENCE

Apr 2024 - Current

ML Research Scientist

Liquid AI (MIT), Boston, United States

Member of technical staff: core research and engineering team. Working on new-generation foundational models. Tasks ranging all the way from machine learning research, model fine-tuning, and explainability/safety to customer use-cases, deployment, and scalability.

Aug 2020 - Current

XAI & NLP Academic Researcher

Technical University of Munich, Munich, Germany

Focus on post-hoc explainability for NLP to improve model interpretability, robustness, and human controllability. Led 20+ ML projects, authored 10+ scientific publications (@ ACL, COLING, MDPI, ECML, etc.). **Teaching (Master Level):** (Advanced) Natural Language Processing | Practical Courses - NLP Applications, XAI for Machine Learning.

Jan 2022 - Dec 2023

NLP Project Lead

Software Campus Program, Munich, Germany

I was selected by the German Ministry for research, which awarded me 100.000€ to lead and develop a project under the mentorship of a major German industry partner: the Holtzbrinck Publishing Group. Led a team of ~10 people to develop a platform for the interactive and controllable deployment of NLP models.

Sep 2022 - Aug 2023

Co-Founder and CTO

Cryptoscape.ai, Berlin, Germany

I led the development of an NLP-powered platform for novice investors to quickly grasp blockchain projects and concepts. Our system simplifies complex information, promoting blockchain adoption. Our work has been supported from TU Berlin's and the [KIEZ](#) accelerator. Tech Stack: Pytorch, FastAPI, Supabase.

Nov 2018 - Apr 2020

ML Research And Development Assistant

Infineon Technologies, Munich, Germany

I applied computer vision and non-standard representation learning to estimate and optimize SoC hardware design metrics. My work resulted in three scientific publications where I co-authored as the 1st or 2nd author. Tech Stack: Tensorflow, Keras, Python, bash.

Feb 2018 - Oct 2018

Data Scientist and Software Developer

ParkHere, Munich, Germany

Designed, developed, and tested data-driven algorithms for automating pedestrian and car detection in large commercial parking lots, using data from multiple sensors tailored to parking slot types. The output of my work has been deployed in production. Tech Stack: Python (Numpy, pandas, scikit-learn), Git, C++, SQL.

PROGRAMMING SKILLS

Data Science: Python 🐍 (+ Pandas, Numpy, Scikit-learn, Jupyter, etc..) | Mojo 🔥

ML/NLP: Pytorch 🔥 | Hugging Face 🗨️ | LangChain 🗝️ | LMQL | Tensorflow-Keras 🔥

Backend/BaaS: FastAPI ⚡ | Supabase ⚡ | **OS / Version Control:** Linux (bash) 🖥️ | Git 📦

Databases: PostgreSQL 🗄️ (+ pgvector) | MongoDB 🌿

EDUCATION

Aug 2020 – Sep 2023

Doctor in NLP and Explainable AI, Computer Science

Technical University of Munich, Germany

Research in the fields of Explainable AI and NLP. More specifically, my PhD thesis explores the usage of post-hoc explainability to work towards more interpretable, robust, and (human) controllable NLP models.

Dissertation: *Explainable AI for the Human-Centric Development of NLP Models*. **Supervisor:** Prof. Dr. Georg Groh.

Oct 2017 – Apr 2020

M.Sc. Mathematics in Data Science

Technical University of Munich, Germany

Focus on machine learning from both the mathematical (theoretical) and computer science (hands-on) sides. Practical experiences in computer vision and NLP. **Thesis:** *Explainability of Hate Speech Detection Models*.

Sep 2013 – May 2017

B.Sc. Mathematics

University of Pisa, Italy

Minor in computer science. **Thesis:** *Computational Classification for Regular Expression Membership Testing*

SELECTED PUBLICATIONS

ACL
2022

Edoardo Mosca et al. *"That Is a Suspicious Reaction!": Interpreting Logits Variation to Detect NLP Adversarial Attacks.*

COLING
2022

Edoardo Mosca et al. *SHAP-Based Explanation Methods: A Review for NLP Interpretability.*

AACL
2023

Edoardo Mosca et al. *IFAN: An Explainability-Focused Interaction Framework for Humans and NLP Models*

MDPI
2023

Edoardo Mosca et al. *Distinguishing Fact from Fiction: A Benchmark Dataset for Identifying Machine-Generated Scientific Papers in the LLM Era. (TrustNLP Extension)*

CERTIFIED TRAININGS

○ Agile for Management

IAV, Berlin, 2 days, Aug 2022

○ Customer-Centric Design Thinking

Grundfos, Denmark, 2 days, Sep 2022

Workplace Personalities

○ Merck, Darmstadt, 2 days, Nov 2022

○ Hiring and Talent Acquisition

Zeiss, Oberkochen, 2 days, Nov 2022

○ Change Management

Trumpf, Stuttgart, 2 days, May 2023

Culture & Sustainability

○ Huawei, Munich, 1 day, June 2023

LANGUAGES

Italian (Native) | English (C2) | German (C1) | Spanish (C1) | French (A2)

REFERENCES

Prof. Dr. Georg Groh (TUM)
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Dr. Lorenzo Servadei (Sony AI)
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Dr. Robert Seidl (Stats Perform)
robertseidl425@gmail.com